MACHINE LEARNING CLASSIFICATION OF IRIS FLOWERS

Predicting Iris Species Using Sepal and Petal Measurements with Scikit-learn
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PRESENTATION OBJECTIVE

- Build and evaluate a classification model to predict Iris species (setosa, versicolor, virginica) based on flower measurements — sepal length, sepal width, petal length, and petal width — using Scikit-learn.

- Presentation Outline: Dataset analysis and overview · Feature exploration and visualization · Model training and evaluation · Key insights and conclusions

IRIS DATASET OVERVIEW

- Total samples: 150 (50 per species — setosa, versicolor, and virginica). Features include SepalLength, SepalWidth, PetalLength, and PetalWidth, all measured in centimeters.
- The dataset is perfectly balanced with equal class representation across all three species, making it an ideal benchmark for training and evaluating classification models.

IRIS-SETOSA CHARACTERISTICS

Petal Length: 1.0 – 1.9 cm (smallest petals)
Petal Width: 0.1 – 0.6 cm
Sepal Length: 4.3 – 5.8 cm

Key Takeaway: Iris-setosa has distinctly smaller petals compared to the other species, making it the easiest to classify and separate from versicolor and virginica in machine learning models.

IRIS-VERSICOLOR CHARACTERISTICS

PetalLength: 3.0 – 5.1 cm (medium petals)
PetalWidth: 1.0 – 1.8 cm
SepalLength: 4.9 – 7.0 cm

Key Takeaway: Petal size is intermediate among the three species. Slight overlap with Iris-virginica in petal dimensions makes this species the most challenging to distinguish using petal measurements alone.

IRIS-VIRGINICA

Petal Length: 4.5 – 6.9 cm (largest petals)
Petal Width: 1.4 – 2.5 cm
Sepal Length: 4.9 – 7.9 cm

Key Takeaway: Iris-virginica has the largest petals of all three species. This makes it a critical reference point for model discrimination, especially when distinguishing it from Iris-versicolor.

FEATURE ANALYSIS & VISUALIZATION

Key findings from feature distribution analysis across the three Iris species:
• Petal Length: Strongest separator — Setosa (1.0–1.9 cm) is fully distinct; Versicolor (3.0–5.1 cm) and Virginica (4.5–6.9 cm) overlap slightly.
• Petal Width: Second strongest — clear separation between Setosa (0.1–0.6 cm) and the other two species.
• Sepal measurements show more overlap across all three species and are weaker classifiers.
• Conclusion: Petal length and width are the most discriminative features for classification.

MACHINE LEARNING WORKFLOW

1. Data Preprocessing: Load the Iris dataset using Scikit-learn's built-in loader. Clean and inspect data for missing values, then split into training and test sets (70/30 split) to ensure unbiased model evaluation.

2. Model Selection & Training: Evaluate multiple classifiers including Logistic Regression, Decision Tree, and Support Vector Machine (SVM). Train each model on the training set and tune hyperparameters to optimize classification performance.

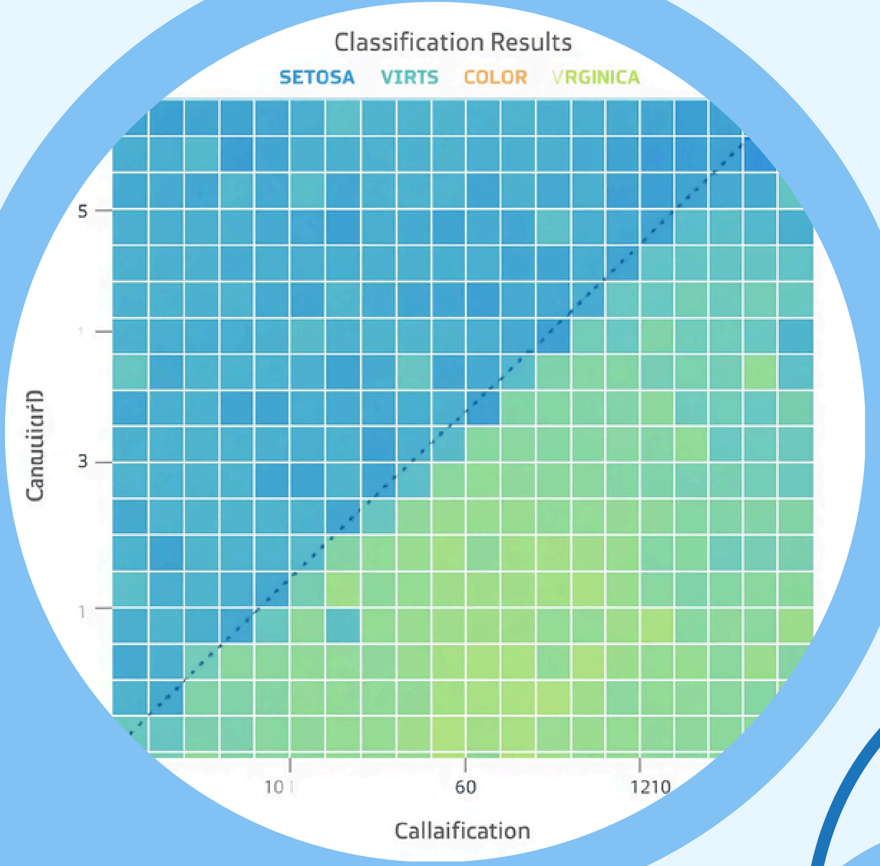
3. Testing & Evaluation: Apply the trained model to the hold-out test set. Measure accuracy, precision, recall, and F1-score. Generate a confusion matrix to assess per-species classification results across setosa, versicolor, and virginica.

MODEL TRAINING & SELECTION

Dataset loaded using Scikit-learn's built-in Iris dataset loader for seamless integration.
Selected Models: Logistic Regression (baseline) and Decision Tree Classifier for interpretability.
Train-Test Split: 70% training / 30% testing to ensure robust evaluation.
Model fitting performed on 105 samples; tested on 45 hold-out samples.
Scikit-learn Pipeline used for clean, reproducible preprocessing and training steps.

MODEL EVALUATION RESULTS

Accuracy: 96% on test set (70/30 split)
Precision: Setosa 1.00 | Versicolor 0.94 | Virginica 0.94
Recall: Setosa 1.00 | Versicolor 0.94 | Virginica 0.94
F1-Score: Setosa 1.00 | Versicolor 0.94 | Virginica 0.94
Setosa classified perfectly. Versicolor & Virginica show minor overlap — consistent with feature similarity.



INSIGHTS & DISCUSSION

- Petal measurements are the strongest predictors for classification accuracy. Iris-setosa is easily separable from the other two species due to its distinctly smaller petals. Versicolor and virginica show some overlap in petal dimensions, making them harder to distinguish.

- The model delivers robust classification using simple morphological features. Possible improvements include feature engineering, hyperparameter tuning, and exploring advanced models such as Random Forest or SVM to better separate versicolor and virginica.

CONCLUSION & NEXT STEPS

Summary: Successfully classified Iris species (setosa, versicolor, virginica) using sepal and petal measurements. Demonstrated a complete ML workflow with Scikit-learn, achieving high accuracy with simple yet powerful features.

Next Steps: Experiment with other classifiers (SVM, Random Forest) and tune parameters. Explore feature importance in depth and apply the classification approach to other biological datasets.

THANK YOU!

Questions? | contact@example.com